

Programming Assignment: Conditional Statements

Part A: Conceptual Foundations

1. What is a conditional statement in programming?

A conditional statement is a control flow structure that allows a program to execute specific blocks of code based on whether a given condition evaluates to true or false. It enables dynamic execution paths rather than a strict sequential flow.

2. What does an if statement do?

An if statement evaluates a target condition or expression. If that expression resolves to true (non-zero), the program transfers control to the nested block of code immediately following the statement. If false, that block is skipped entirely.

3. What is the purpose of an else block?

The purpose of an else block is to provide an explicit alternative execution path. It ensures that if the condition in the preceding if statement evaluates to false, a default group of instructions is guaranteed to run.

4. When is an else block executed?

An else block executes if and only if the primary condition (and all associated intermediate else if conditions) evaluates to false.

5. What is a Boolean expression?

A Boolean expression is an algebraic or relational statement that, when compiled and evaluated, produces a value of either true or false. In traditional C, false is represented by 0 and true is represented by any non-zero value; C++ explicitly uses the **bool** type containing keywords **true** or **false**.

6. Write the syntax of a simple if statement.

The basic syntax in C and C++ requires conditional parentheses and code block encapsulation:

```
if (condition) {  
    // Code to execute if condition is true  
}
```

7. What operator is commonly used to test equality?

The double equals operator (==) is used to test equality. It compares the operands on its left and right sides, returning true if they match, distinct from the single equals operator (=) which is used for value assignment.

8. What does the modulus (%) operator return?

The modulus operator (%) performs integer division between two operands and returns the remainder. For instance, 7 % 3 returns 1.

9. How can you check if a number is even?

A number is even if it leaves no remainder when divided by 2. This is implemented via a relational check against zero using the modulus operator:

```
#include <iostream>

int main() {

    int inputNumber;

    std::cout << "Enter an integer: ";

    std::cin >> inputNumber;

    // A number is even if divided by 2 leaves a remainder of 0

    if (inputNumber % 2 == 0) {

        std::cout << inputNumber << " is an EVEN number.\n";

    } else {

        std::cout << inputNumber << " is an ODD number.\n";

    }

    return 0;

}
```

10. How can you check if a number is odd?

A number is odd if it yields a remainder of 1 (or a non-zero value) when divided by 2. It can be checked in either of the following formats:

```
#include <iostream>
```

```
int main() {
```

```

double numericValue;

std::cout << "Enter any number: ";

std::cin >> numericValue;

// Sequential if-else-if ladder to check signs

if (numericValue > 0) {

    std::cout << "The value is POSITIVE.\n";

} else if (numericValue < 0) {

    std::cout << "The value is NEGATIVE.\n";

} else {

    std::cout << "The value is ZERO.\n";

}

return 0;

}

```

11. What is the purpose of else-if?

The else if statement allows the programmer to link multiple mutually exclusive conditions together. It enables the evaluation of subsequent expressions sequentially until one is found true, bypassing a heavy nesting structure.

12. What is a nested if statement?

A nested if statement occurs when an if or else if block is placed inside the body of another preceding if block. It allows for multi-tier criteria testing, where the secondary condition is only checked if the outer condition succeeds.

13. Can an if statement exist without an else?

Yes. An if statement is entirely autonomous. The inclusion of an else block is optional and is only added when alternative actions need to be systematically handled upon conditional failure.

Part B: Operators and Logical Evaluation

14. What is the output when a false condition is tested in an if statement without else?

No code within the if structure is executed. The program simply passes over the entire condition block and skips straight to the next sequential statement listed outside the brackets.

15. What is the role of braces {} in C/C++ conditionals?

Braces define a compound block statement. They group multiple logical lines of code together under a single control directive. While single statements can technically omit braces, using them prevents logical tracking bugs and ensures clarity.

16. What does > mean?

The strict "Greater Than" relational operator. It returns true if the value on the left-hand side is quantitatively larger than the value on the right-hand side.

17. What does < mean?

The strict "Less Than" relational operator. It returns true if the left-hand side value is smaller than the right-hand side value.

18. What does >= mean?

The "Greater Than or Equal To" relational operator. It returns true if the left-hand value is either larger than or equal to the right-hand value.

19. What does <= mean?

The "Less Than or Equal To" relational operator. It yields true if the left-hand value is less than or identical to the right-hand value.

20. What does != mean?

The "Not Equal To" inequality operator. It returns true if the values on both sides do not match, and false if they are identical.

21. What is a switch statement?

A switch statement is a multi-way branch selection control structure. It evaluates an expression (yielding an integer or character type) and matches its result against a series of defined constant options called "cases," routing control directly to the matching block.

22. What is a case label in a switch statement?

A case label is an explicit constant value indicator mapped within a switch statement block. If the value of the main expression matches this constant value, program execution jumps directly to the statements following that specific label.

23. Why is break used in switch?

The break keyword terminates the active switch structure instantly, skipping all remaining execution paths and forcing control out of the block. Without a break, the execution path bleeds downward into sequential cases.

24. What happens if break is omitted?

If a break is omitted, execution shifts linearly into the subsequent case blocks below it regardless of whether those labels match the conditional expression. This behavior is referred to as "fall-through."

25. What is the default case in switch?

The default case acts as a catch-all block inside a switch statement. It executes if none of the explicit case labels match the value of the evaluated conditional expression. It serves the same architectural purpose as a final else statement in an if-else ladder.

26. Can switch compare strings in C++?

No, native C++ switch structures cannot evaluate strings. They only operate on integral or enumeration types (such as int, char, or enum). To check strings, you must use an if-else ladder combined with comparison operators or hash maps.

27. Which statement is better for many constant choices: if-else or switch?

The switch statement is structurally better for managing a large collection of constant values. Compilers optimize switch statements using a mechanism called a "jump table," which provides highly efficient $O(1)$ evaluation speeds compared to the sequential $O(n)$ scanning behavior of if-else ladders.

Part C: Structural Implementation and Logic Patterns

28. How do you test whether a value is positive?

A value is positive if it is quantitatively greater than zero:

```
if (value > 0) { /* Value is positive */ }
```

29. How do you test whether a value is negative?

A value is negative if it is strictly less than zero:

```
if (value < 0) { /* Value is negative */ }
```

30. How do you test whether a value is zero?

A value is explicitly tested against zero using the equality operator:

```
if (value == 0) { /* Value is zero */ }
```

31. Write a condition to check if age is at least 18.

To verify an age is 18 or older, use the greater-than-or-equal-to operator:

```
if (age >= 18) { /* Person is an adult */ }
```

32. Write a condition to check if marks are above 50.

To evaluate if score values are strictly higher than 50:

```
if (marks > 50) { /* Marks are above 50 */ }
```

33. What is decision making in programming?

Decision making is the fundamental design concept where a system evaluates structural states or environment variables during runtime execution to determine an optimal programmatic step, branching execution paths according to strict logical criteria.

34. Can nested if statements have multiple levels?

Yes. Nested if structures can theoretically scale to arbitrary depth levels. However, stacking nesting past 3 to 4 levels degrades readability and increases cognitive load, which is often called the "arrow anti-pattern."

35. What is logical AND (&&)?

The logical AND operator (&&) combines two boolean sub-expressions. It yields true if and only if both conditions evaluate to true. If the left condition evaluates to false, it skips evaluating the right condition entirely due to a process called "short-circuit evaluation."

36. What is logical OR (||)?

The logical OR operator (||) connects two conditions and evaluates to true if at least one of them is true. It also features short-circuit evaluation: if the left-side expression is verified true, the right-hand condition is bypassed completely since the global truth is already established.

37. What is logical NOT (!)?

The logical NOT operator (!) is a unary operator that inverts the boolean state of its operand. It transforms a true value into false, and a false value into true.

38. How do you check if a year is divisible by 4?

By applying the modulus operator against 4 and checking for a zero remainder:

```
if (year % 4 == 0) {
```

```
    // True: The year is cleanly divisible by 4
```

```
}
```

39. What is a leap year?

A leap year is an extended calendar year containing 366 days instead of 365. This adjustment happens once every four years to align the calendar year with the astronomical solar year.

40. What is the first condition for a leap year?

The primary condition is that the target year must be cleanly divisible by 4. However, to account for exceptions, centurial years (ending in 00) must also be divisible by 400 to be classified as leap years.

```
year % 4 == 0
```

Part D: Advanced Selection and Problem Analysis

41. How do you find the largest of two numbers?

```
#include <iostream>

int main() {

    double number1, number2;

    // Request inputs from the user

    std::cout << "Enter the first number: ";

    std::cin >> number1;

    std::cout << "Enter the second number: ";

    std::cin >> number2;


    // Evaluation logic using if-else-if ladder

    if (number1 > number2) {

        std::cout << "The largest number is: " << number1 << "\n";

    }

    else if (number2 > number1) {

        std::cout << "The largest number is: " << number2 << "\n";

    }

}
```

```

else {

    std::cout << "Both numbers are completely equal.\n";

}

return 0;

}

```

42. How do you find the largest of three numbers?

```

#include <iostream>

int main() {

    double num1, num2, num3;

    std::cout << "Enter three numbers separated by spaces: ";

    std::cin >> num1 >> num2 >> num3;

    // Finding the largest of two numbers (Comparing num1 and num2)

    std::cout << "\n--- Comparison (First Two Numbers) ---\n";

    if (num1 > num2) {

        std::cout << num1 << " is larger than " << num2 << "\n";

    } else if (num2 > num1) {

        std::cout << num2 << " is larger than " << num1 << "\n";

    } else {

        std::cout << "Both numbers are completely equal.\n";

    }

    // Finding the largest of all three numbers using logical AND (&&)

    std::cout << "\n--- Comparison (All Three Numbers) ---\n";

    double maximumValue;

    if (num1 >= num2 && num1 >= num3) {

```

```

    maximumValue = num1;

} else if (num2 >= num1 && num2 >= num3) {

    maximumValue = num2;

} else {

    maximumValue = num3;

}

std::cout << "The absolute largest value among the three is: " << maximumValue << "\n";

return 0;

}

```

43. What is an if-else-if ladder?

An if-else-if ladder is a common programming pattern used to evaluate multiple sequential conditions. The program checks conditions from top to bottom; as soon as a condition evaluates to true, its corresponding code block runs, and the rest of the ladder is skipped.

44. What is the difference between if and switch?

An if statement handles variable ranges, compound logical expressions (using && or ||), and complex data conditions. A switch statement is restricted to checking single, discrete constant values of integral or character types.

45. Can switch use ranges directly?

Standard ISO C/C++ does not support defining numeric range bounds (e.g., case 10...20:) directly inside case labels. Each individual value must have its own distinct case label, or you must use an if-else-if structure to evaluate ranges instead.

46. What is fall-through in switch?

Fall-through is a behavior in a switch block where execution continues into subsequent cases because a break statement was omitted. While sometimes intentional, it is often an accidental bug. Modern compilers often require developers to add explicit annotations (like `[[fallthrough]]`;) to confirm it is intentional.

47. What data type is usually used for conditions?

Boolean data types (bool in C++) are typically used for conditions. In standard C applications, integer types (int) are used instead, where zero evaluates as false and any non-zero value evaluates as true.

48. Why are conditional statements important?

Without conditional logic, programs would execute linearly from the first line to the last, producing identical results every run regardless of user input. Conditional structures give applications the flexibility to react, specialize, validate inputs, handle errors, and make real-time decisions.

49. Give one real-life example of a conditional statement.

An automated teller machine (ATM) workflow uses conditional logic: **IF** a user's requested withdrawal amount is less than or equal to their current account balance, the machine dispenses the cash. **ELSE**, it denies the transaction and displays an "Insufficient Funds" error message.

50. Name four conditional structures covered.

The four primary conditional structures used to control program flow are:

1. Simple **if** statements
2. Standard **if-else** blocks
3. Sequential **if-else-if** ladders
4. Discrete multi-branch **switch** blocks